Implementing DES Encryption Algorithm in Verilog

- Presented by: Roni Zukerman and David Orlian
- Course: Cryptography Algorithms and Verilog Implementation (130330)
- Institution: Jerusalem College of Technology

- Date: 21/9/2025

The DES Encryption Algorithm

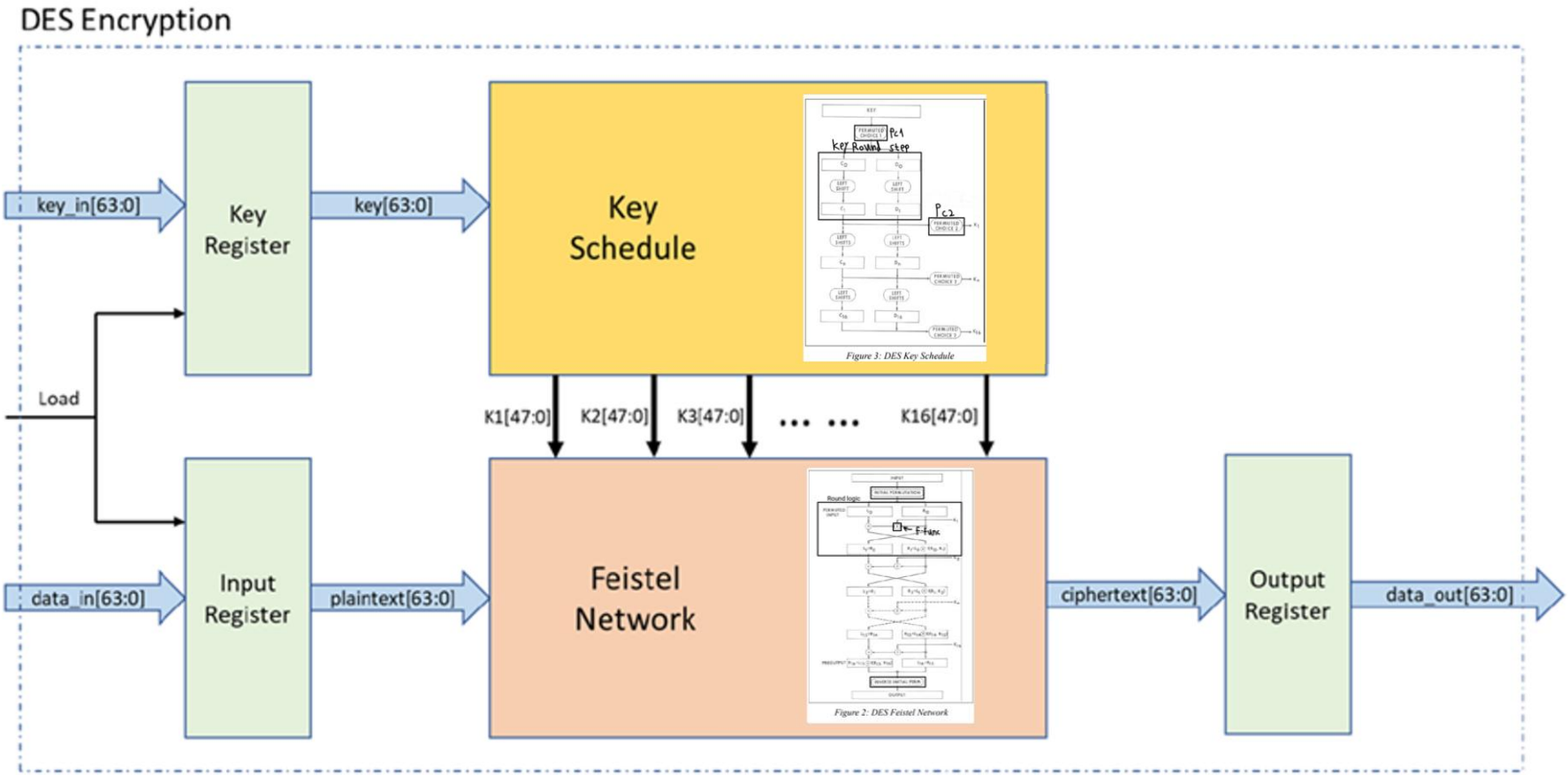


Figure 1: DES Encryption, Block Diagram

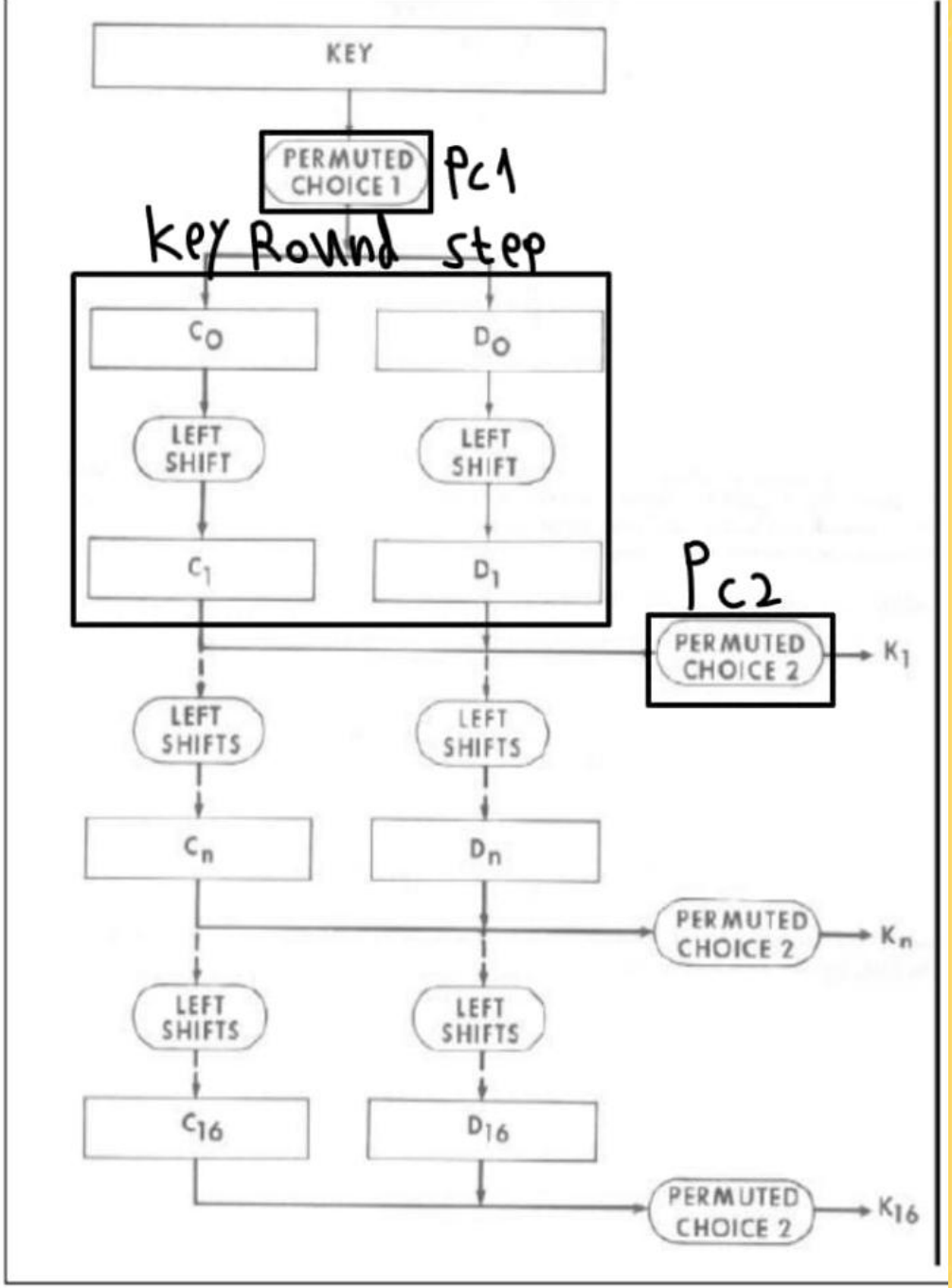


Figure 3: DES Key Schedule

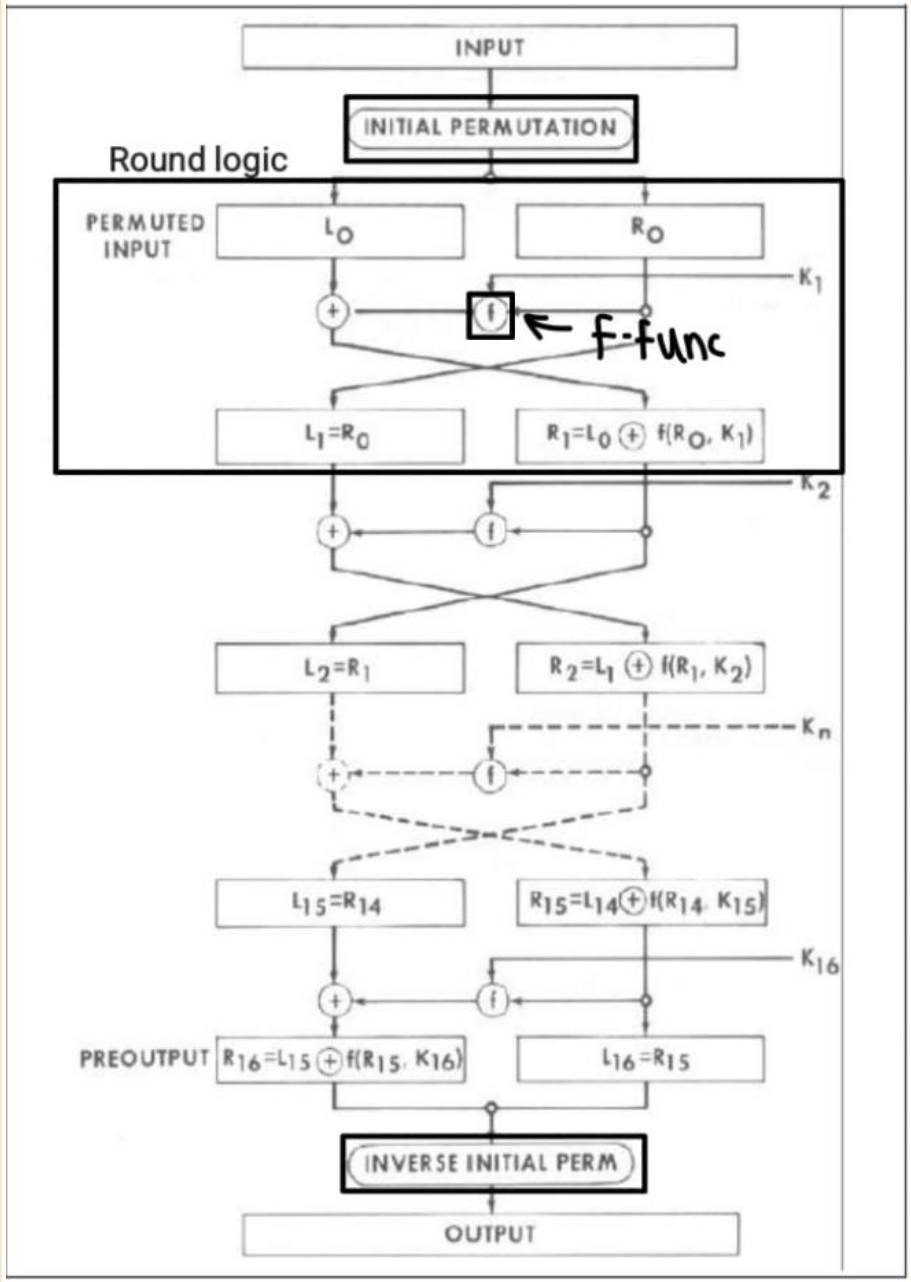
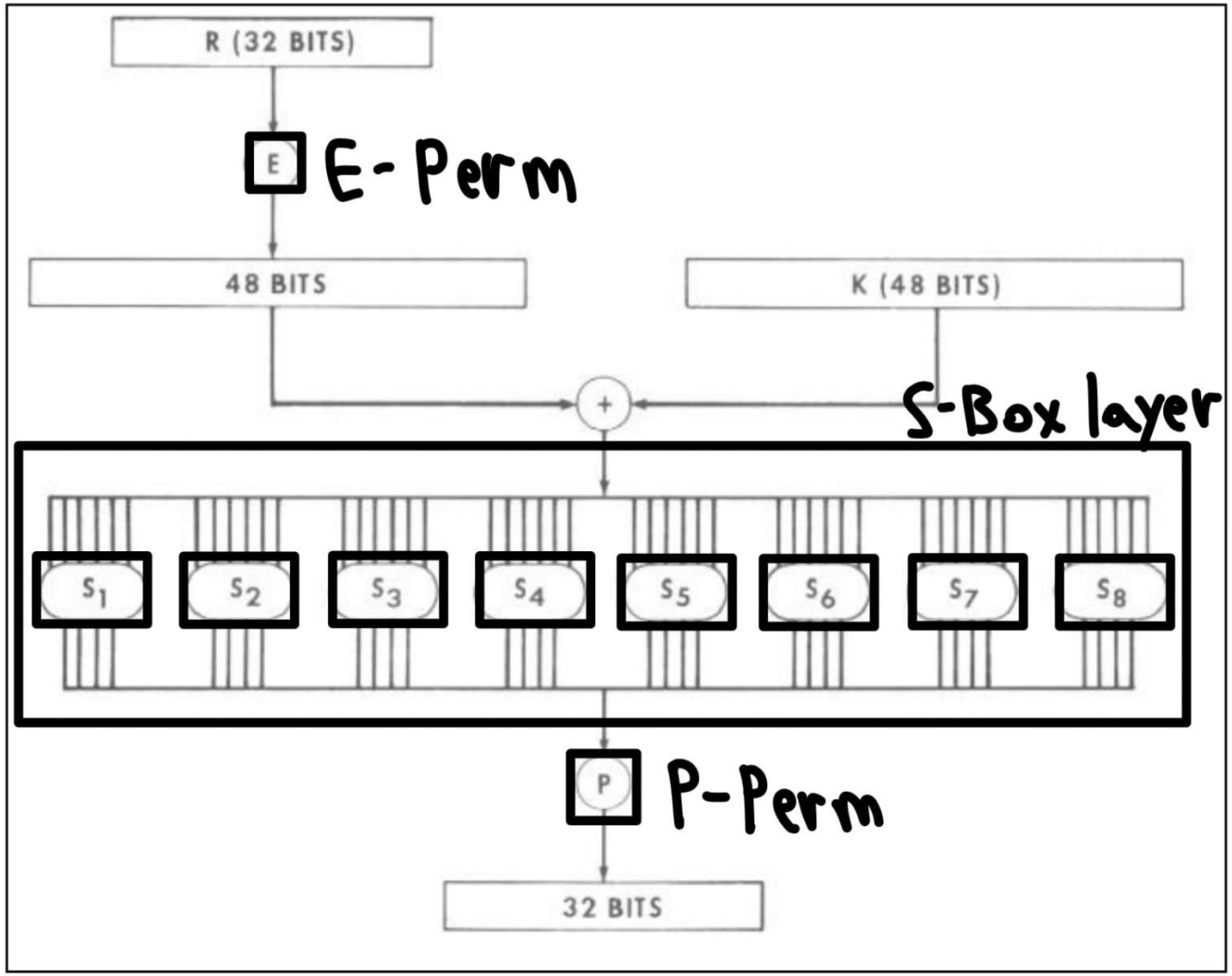
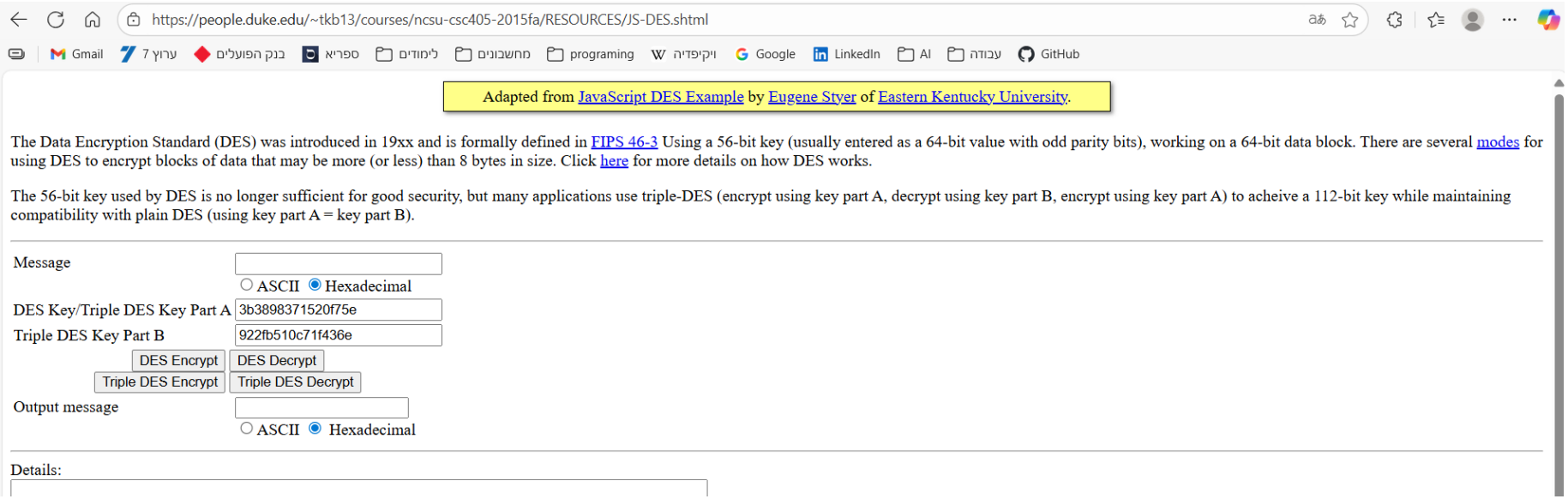


Figure 2: DES Feistel Network


R (32 BITS)
E
E-Perm
48 BITS
K (48 BITS)
+
S-Box layer
S1
S2
S3
S4
S5
S6
S7
S8
P
P-Perm
32 BITS

Testbench

- Most of the testbenches are in the form of:
  input data, output data and expected output
- Python script to generate reliable test vectors with:
  https://people.duke.edu/~tkb13/courses/ncsu-csc405-2015fa/RESOURCES/JS-DES.shtml

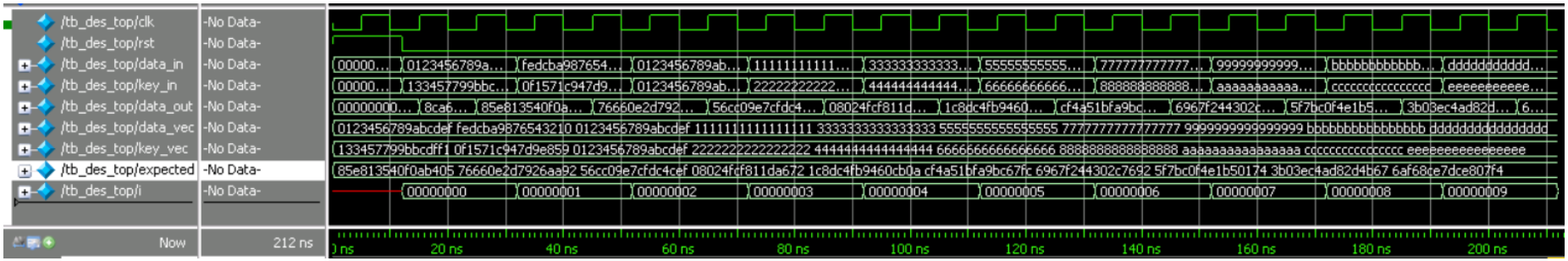


Simulation Results

Transcript:

```
VSIM 27> run 1 us
# Test  1 | data_in=0123456789abcdef | key_in=133457799bbcdff1 | data_out=85e813540f0ab405 | expected=85e813540f0ab405
# Test  1 PASS
# Test  2 | data_in=fedcba9876543210 | key_in=0f1571c947d9e859 | data_out=76660e2d7926aa92 | expected=76660e2d7926aa92
# Test  2 PASS
# Test  3 | data_in=0123456789abcdef | key_in=0123456789abcdef | data_out=56cc09e7cfdc4cef | expected=56cc09e7cfdc4cef
# Test  3 PASS
# Test  4 | data_in=1111111111111111 | key_in=2222222222222222 | data_out=08024fcf811da672 | expected=08024fcf811da672
# Test  4 PASS
# Test  5 | data_in=3333333333333333 | key_in=4444444444444444 | data_out=1c8dc4fb9460cb0a | expected=1c8dc4fb9460cb0a
# Test  5 PASS
# Test  6 | data_in=5555555555555555 | key_in=6666666666666666 | data_out=cf4a51bfa9bc67fc | expected=cf4a51bfa9bc67fc
# Test  6 PASS
# Test  7 | data_in=7777777777777777 | key_in=8888888888888888 | data_out=6967f244302c7692 | expected=6967f244302c7692
# Test  7 PASS
# Test  8 | data_in=9999999999999999 | key_in=aaaaaaaaaaaaaaaa | data_out=5f7bc0f4e1b50174 | expected=5f7bc0f4e1b50174
# Test  8 PASS
# Test  9 | data_in=bbbbbbbbbbbbbbbb | key_in=cccccccccccccccc | data_out=3b03ec4ad82d4b67 | expected=3b03ec4ad82d4b67
# Test  9 PASS
# Test 10 | data_in=dddddddddddddddd | key_in=eeeeeeeeeeeeeeee | data_out=6af68ce7dce807f4 | expected=6af68ce7dce807f4
# Test 10 PASS
# ** Note: $finish    : tb_des_top.v(74)
#    Time: 212 ns  Iteration: 0  Instance: /tb_des_top
# 1
# Break in Module tb_des_top at tb_des_top.v line 74
```

Waveform:



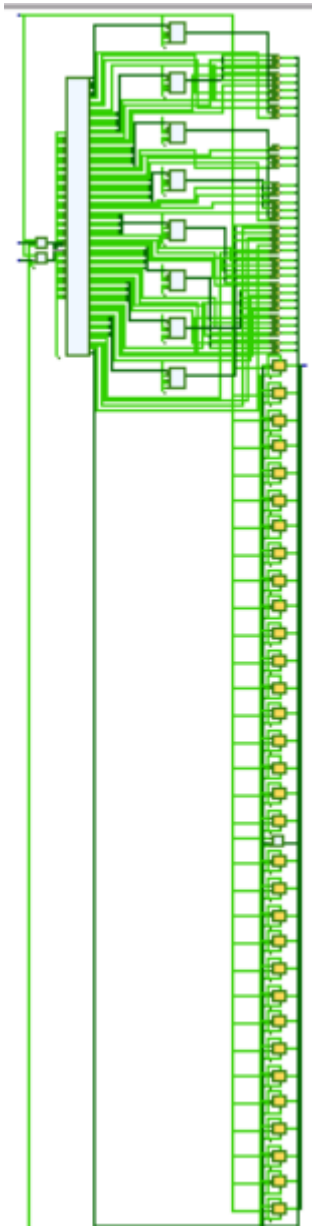
Project Architecture

- Approach: Loop unrolling architecture.
- Target Device: Xilinx Artix-7 FPGA.
- Frequency: Slowed down for implementation to 10MHz.

Synthesis Results

- Tool Used: Vivado/Precision
- Summary: Area utilization, critical path delay.
- Screenshot: Synthesis summary report.

RTL schematic:



Technology schematic:

Place & Route Results

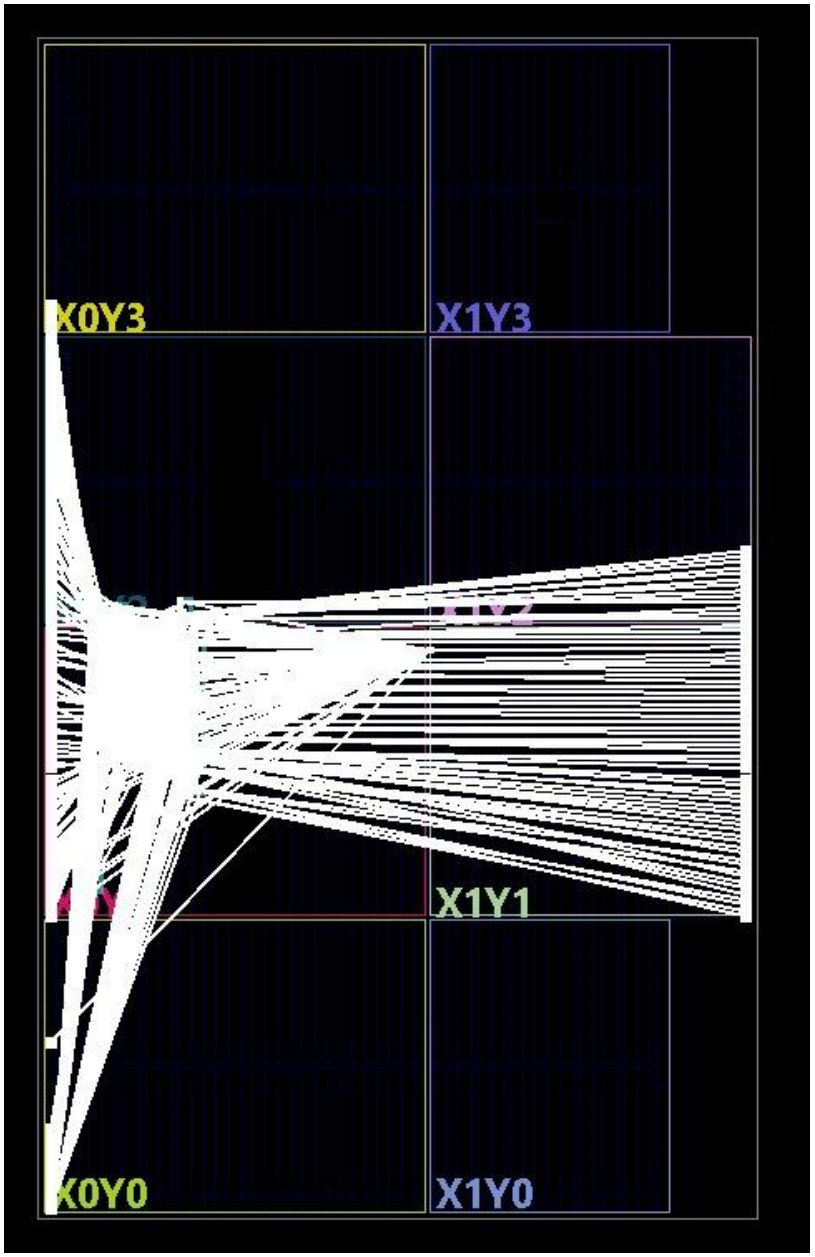
- Screenshots: Place & Route overview.
- Timing Reports: Achieved timing constraints.
- Utilization: Final device resource usage.

```
+-------------------------+------+-------+-----------+-------+
|        Site Type        | Used | Fixed | Available | Util% |
+-------------------------+------+-------+-----------+-------+
| Slice LUTs              | 1151 |     0 |     47200 |  2.44 |
|   LUT as Logic          | 1151 |     0 |     47200 |  2.44 |
|   LUT as Memory         |    0 |     0 |     19000 |  0.00 |
| Slice Registers         |  184 |     0 |     94400 |  0.19 |
|   Register as Flip Flop |  184 |     0 |     94400 |  0.19 |
|   Register as Latch     |    0 |     0 |     94400 |  0.00 |
| F7 Muxes                |    0 |     0 |     31700 |  0.00 |
| F8 Muxes                |    0 |     0 |     15850 |  0.00 |
+-------------------------+------+-------+-----------+-------+
```

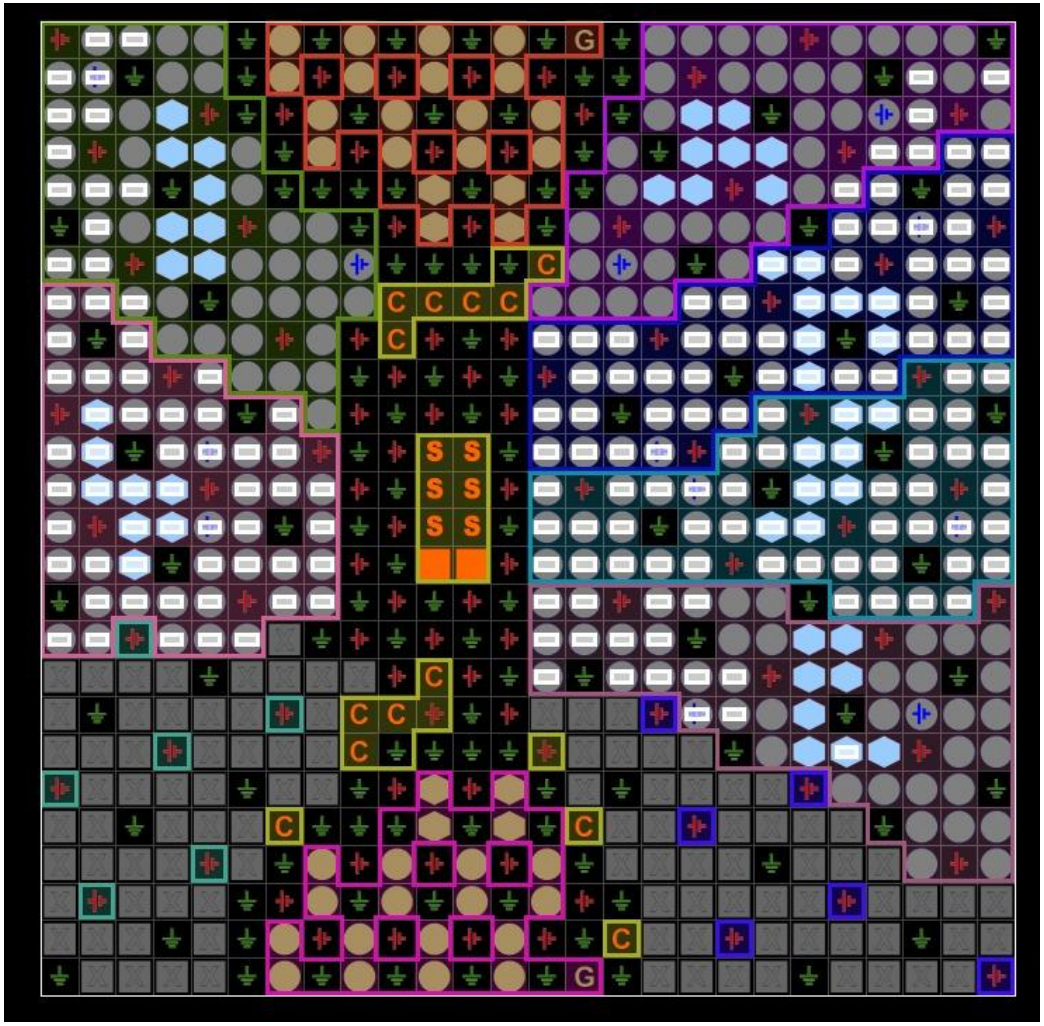
I/O Reports

- Pin Mapping: Constraints applied.
- I/O Report: I/O utilization summary.
- Screenshot: I/O constraints file or Vivado I/O view.

Device layout:



I/O report



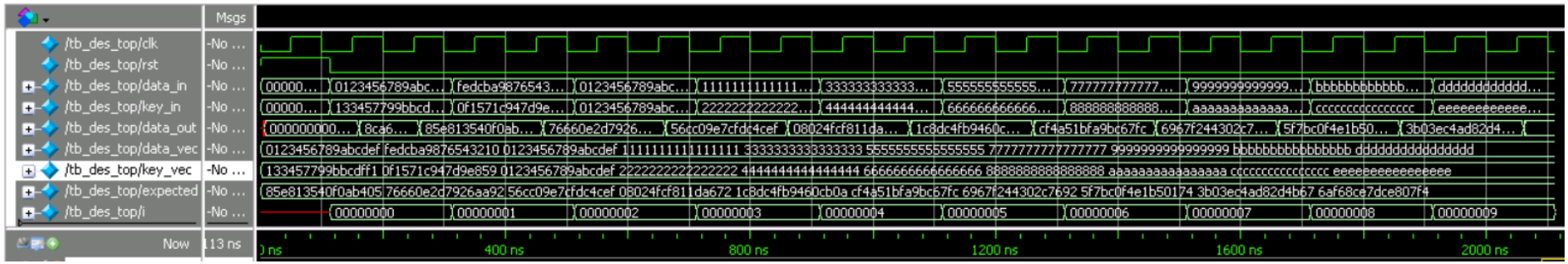
Source & Gate-Level Simulation Results

- Tools: ModelSim
- Screenshots: Sample waveforms.
- Outputs: $display and $monitor for debugging.
- Validation: Test vectors matched expected DES outputs.

Transcript:

```
VSIM 10> run 10 us
# Test  1 | data_in=0123456789abcdef | key_in=133457799bbcdff1 | data_out=85e813540f0ab405 | expected=85e813540f0ab405
# Test  1 PASS
# Test  2 | data_in=fedcba9876543210 | key_in=0f1571c947d9e859 | data_out=76660e2d7926aa92 | expected=76660e2d7926aa92
# Test  2 PASS
# Test  3 | data_in=0123456789abcdef | key_in=0123456789abcdef | data_out=56cc09e7cfdc4cef | expected=56cc09e7cfdc4cef
# Test  3 PASS
# Test  4 | data_in=1111111111111111 | key_in=2222222222222222 | data_out=08024fcf811da672 | expected=08024fcf811da672
# Test  4 PASS
# Test  5 | data_in=3333333333333333 | key_in=4444444444444444 | data_out=1c8dc4fb9460cb0a | expected=1c8dc4fb9460cb0a
# Test  5 PASS
# Test  6 | data_in=5555555555555555 | key_in=6666666666666666 | data_out=cf4a51bfa9bc67fc | expected=cf4a51bfa9bc67fc
# Test  6 PASS
# Test  7 | data_in=7777777777777777 | key_in=8888888888888888 | data_out=6967f244302c7692 | expected=6967f244302c7692
# Test  7 PASS
# Test  8 | data_in=9999999999999999 | key_in=aaaaaaaaaaaaaaaa | data_out=5f7bc0f4e1b50174 | expected=5f7bc0f4e1b50174
# Test  8 PASS
# Test  9 | data_in=bbbbbbbbbbbbbbbb | key_in=cccccccccccccccc | data_out=3b03ec4ad82d4b67 | expected=3b03ec4ad82d4b67
# Test  9 PASS
# Test 10 | data_in=dddddddddddddddd | key_in=eeeeeeeeeeeeeeee | data_out=6af68ce7dce807f4 | expected=6af68ce7dce807f4
# Test 10 PASS
# ** Note: $finish    : tb_des_top.v(74)
#    Time: 2113 ns  Iteration: 0  Instance: /tb_des_top
# 1
# Break in Module tb_des_top at tb_des_top.v line 74
```

Waveform:



Critical Path & Throughput

Before fixing timing issue:

Tcl Console | Messages | Log | Reports | Design Runs | Power | DRC | Methodology | Timing

Design Timing Summary

- General Information
- Timer Settings
- Design Timing Summary
- Clock Summary (1)
- Check Timing (0)
- Intra-Clock Paths
- Inter-Clock Paths
- Other Path Groups

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	22.414 ns	Worst Hold Slack (WHS):	0.164 ns	Worst Pulse Width Slack (WPWS):	4.500 ns
Total Negative Slack (TNS):	1531.404 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	128	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	494	Total Number of Endpoints:	494	Total Number of Endpoints:	216

Timing constraints are not met.

Timing Summary - impl_1 (saved) | Timing Summary - timing_1

After fixing timing issue:

Tcl Console | Messages | Log | Reports | Design Runs | Power | DRC | Methodology | Timing

Design Timing Summary

- General Information
- Timer Settings
- Design Timing Summary
- Clock Summary (1)
- Check Timing (0)
- Intra-Clock Paths
- Inter-Clock Paths
- Other Path Groups

Setup		Hold		Pulse Width	
Worst Negative Slack (WNS):	53.490 ns	Worst Hold Slack (WHS):	0.052 ns	Worst Pulse Width Slack (WPWS):	49.500 ns
Total Negative Slack (TNS):	0.000 ns	Total Hold Slack (THS):	0.000 ns	Total Pulse Width Negative Slack (TPWS):	0.000 ns
Number of Failing Endpoints:	0	Number of Failing Endpoints:	0	Number of Failing Endpoints:	0
Total Number of Endpoints:	432	Total Number of Endpoints:	432	Total Number of Endpoints:	185

All user specified timing constraints are met.

Timing Summary - impl_1 (saved)

Calculation of Fmax and throughput by the vivado results:

$$WNS = 53.49[ns]$$

$$Critical\ Path\ Delay = 100[ns] - 53.49[ns] = \mathbf{46.51}[\boldsymbol{ns}]$$

$$F_{max} = \frac{1}{Critical\ Path\ Delay} = \mathbf{21.5}[\boldsymbol{MHz}]$$

$$Throughput = 64 \times F_{max} = \mathbf{1.376}[\frac{\boldsymbol{GB}}{\boldsymbol{s}}]$$

Conclusions & Lessons Learned

- Challenges: Debugging loop unrolling, simulation vs. synthesis.
- What worked: Design met functional and timing goals.
- Improvements: Possible pipelining, optimization.
- Final note: Thank you and open to questions!